V3 moneyness performance

Same 6 models · 3 companies · 12 windows · 1-day hourly study · classified by call moneyness m = S/K

What this report is

- Companion to V3_1d_hourly_empirical_study. The notebook in this folder is the source of truth; this PDF is written from the same moneyness payload.
- Question: does model ranking change with option moneyness, overall and inside each volatility regime?
- BEST in every table is the lowest RMSE%. That row is bold and green, and the Mark column says BEST. Empty buckets (n = 0) show em dashes and have no BEST.

Unchanged experimental design

- Same six models, three companies (SPY, AAPL, MSFT), 12 windows, 1-day lookback, hourly recalibration, V3 filters, Euler-corrected Heston step, LSM n_paths = 10000 / seed 42.
- Same option contracts as the main study (nearest-ATM listed calls, DTE 7–60, $|S/K - 1| \le 10\%$). No model-specific re-sampling.
- Pooled n = 216 contracts per model (identical keys across models).

Moneyness rule (calls, m = S / K)

- DOTM: $m < 0.98$. OTM: $0.98 \le m < 0.995$. ATM: $0.995 \le m \le 1.005$. ITM: $1.005 < m \le 1.02$. DITM: $m > 1.02$.
- These five bins partition the main-study filter $|m - 1| \le 10\%$. The Monday nearest-ATM design puts most mass in ATM; DOTM and DITM are sparse and appear mainly in 2008–2009.

Metrics

- RMSE% = $100 \times \sqrt{\text{mean}(((C_{model} - C_{mkt})/C_{mkt})^2)}$. Ranking uses this only.
- MAE and Bias $ in option-premium dollars. Bias% = $100 \times \text{mean}((C_{model} - C_{mkt})/C_{mkt})$. Early-exercise fraction = mean LSM share of paths that stop before expiry.

Two analyses

- Analysis 1 — Overall moneyness: pool all companies and regimes. Five tables (one per bucket) plus a Model × Moneyness RMSE% heatmap.
- Analysis 2 — Window × moneyness: keep the 12 windows separate, pool the three companies. Compact RMSE tables/heatmaps and a winner grid showing whether the best model changes with moneyness inside each window.

Page 2 · Shared sample by moneyness · identical for all six models

Table M0 · Contract counts (one model = all models)

Regime	DOTM	OTM	ATM	ITM	DITM	Total
2022-10-21 Oct 2022	0	0	1	5	12	18
2022-11-18 Nov 2022	13	5	0	0	0	18
2022-12-16 Dec 2022	6	1	3	3	5	18
2023-01-20 Jan 2023	12	0	0	1	5	18
2023-02-17 Feb 2023	12	6	0	0	0	18
2023-03-17 Mar 2023	2	8	7	1	0	18
2023-04-21 Apr 2023	5	9	4	0	0	18
2023-05-19 May 2023	18	0	0	0	0	18
2023-06-16 Jun 2023	18	0	0	0	0	18
2023-07-21 Jul 2023	18	0	0	0	0	18
2023-08-18 Aug 2023	18	0	0	0	0	18
2023-09-15 Sep 2023	18	0	0	0	0	18
All regimes	140	29	15	10	22	216

Moneyness rule · American calls, m = S/K

Bucket	Rule	n (pooled)	Share
DOTM	$m < 0.98$	140	64.8%
OTM	$0.98 \le m < 0.995$	29	13.4%
ATM	$0.995 \le m \le 1.005$	15	6.9%
ITM	$1.005 < m \le 1.02$	10	4.6%
DITM	$m > 1.02$	22	10.2%

ATM dominates because the main study picks the nearest-ATM listed call each Monday. DOTM/DITM are not empty because of a new filter — they are rare in that ATM sample, especially after 2009. Every model still sees the same contracts in every bucket.

Page 3 · Analysis 1 · Overall moneyness · Tables M1–M3

Pooled SPY + AAPL + MSFT and all 12 windows. Bold green = lowest RMSE%.

Table M1 · DOTM · m < 0.98 · n = 140 · BEST = Heston

Model	n	RMSE%	MAE	Bias $	Bias%	Early-ex.	Mark
GBM	140	99.93	6.7920	-6.7920	-99.93	0.2%	
GARCH	140	99.93	6.7918	-6.7918	-99.93	0.2%	
Heston	**140**	**99.72**	**6.7777**	**-6.7777**	**-99.71**	**0.5%**	**BEST**
Merton	140	99.93	6.7920	-6.7920	-99.93	0.2%	
GARCH–Merton	140	99.92	6.7911	-6.7911	-99.92	0.3%	
Heston–Merton	140	99.73	6.7785	-6.7785	-99.72	0.5%	

Table M2 · OTM · 0.98 ≤ m < 0.995 · n = 29 · BEST = Heston–Merton

Model	n	RMSE%	MAE	Bias $	Bias%	Early-ex.	Mark
GBM	29	97.46	6.6450	-6.6450	-97.41	7.4%	
GARCH	29	97.69	6.6613	-6.6613	-97.62	5.9%	
Heston	29	94.78	6.4819	-6.4819	-94.58	11.1%	
Merton	29	97.43	6.6436	-6.6436	-97.38	7.8%	
GARCH–Merton	29	97.40	6.6423	-6.6423	-97.32	6.7%	
Heston–Merton	**29**	**94.63**	**6.4714**	**-6.4714**	**-94.41**	**10.3%**	**BEST**

Table M3 · ATM · 0.995 ≤ m ≤ 1.005 · n = 15 · BEST = Heston–Merton

Model	n	RMSE%	MAE	Bias $	Bias%	Early-ex.	Mark
GBM	15	87.74	5.5528	-5.5528	-87.43	30.8%	
GARCH	15	88.07	5.5814	-5.5814	-87.66	29.9%	
Heston	15	83.40	5.2667	-5.2667	-82.78	34.5%	
Merton	15	87.81	5.5584	-5.5584	-87.50	31.9%	
GARCH–Merton	15	87.28	5.5230	-5.5230	-86.85	30.6%	
Heston–Merton	**15**	**83.35**	**5.2589**	**-5.2589**	**-82.71**	**32.6%**	**BEST**

Bias $ = mean(model − market). Bias% = 100 × mean((model − market)/market).

Page 4 · Analysis 1 · Overall moneyness · Tables M4–M5

Pooled SPY + AAPL + MSFT and all 12 windows. Bold green = lowest RMSE%.

Table M4 · ITM · 1.005 < m ≤ 1.02 · n = 10 · BEST = Heston

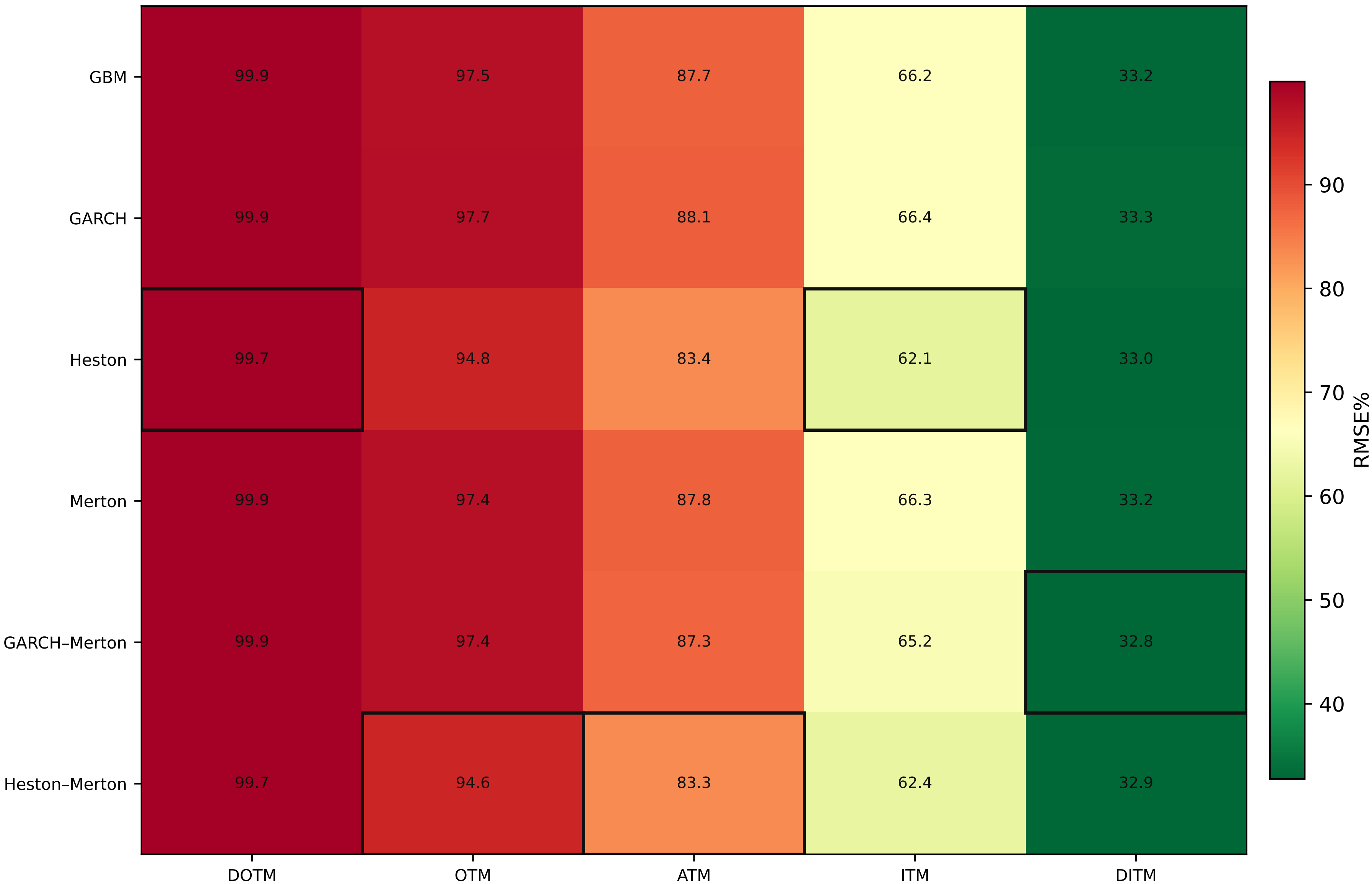
Model	n	RMSE%	MAE	Bias $	Bias%	Early-ex.	Mark
GBM	10	66.15	4.5714	-4.5714	-64.47	64.9%	
GARCH	10	66.45	4.6035	-4.6035	-64.84	66.9%	
Heston	**10**	**62.06**	**4.3125**	**-4.3125**	**-60.05**	**61.0%**	**BEST**
Merton	10	66.26	4.5791	-4.5791	-64.60	66.6%	
GARCH–Merton	10	65.24	4.5263	-4.5263	-63.63	65.5%	
Heston–Merton	10	62.44	4.3385	-4.3385	-60.41	60.1%	

Table M5 · DITM · m > 1.02 · n = 22 · BEST = GARCH–Merton

Model	n	RMSE%	MAE	Bias $	Bias%	Early-ex.	Mark
GBM	22	33.21	3.1364	-3.1364	-28.86	91.7%	
GARCH	22	33.31	3.1521	-3.1521	-29.03	94.6%	
Heston	22	32.95	3.1073	-3.1073	-28.24	86.1%	
Merton	22	33.15	3.1291	-3.1291	-28.76	92.8%	
GARCH–Merton	**22**	**32.77**	**3.0972**	**-3.0972**	**-28.54**	**92.2%**	**BEST**
Heston–Merton	22	32.94	3.1016	-3.1016	-28.18	87.8%	

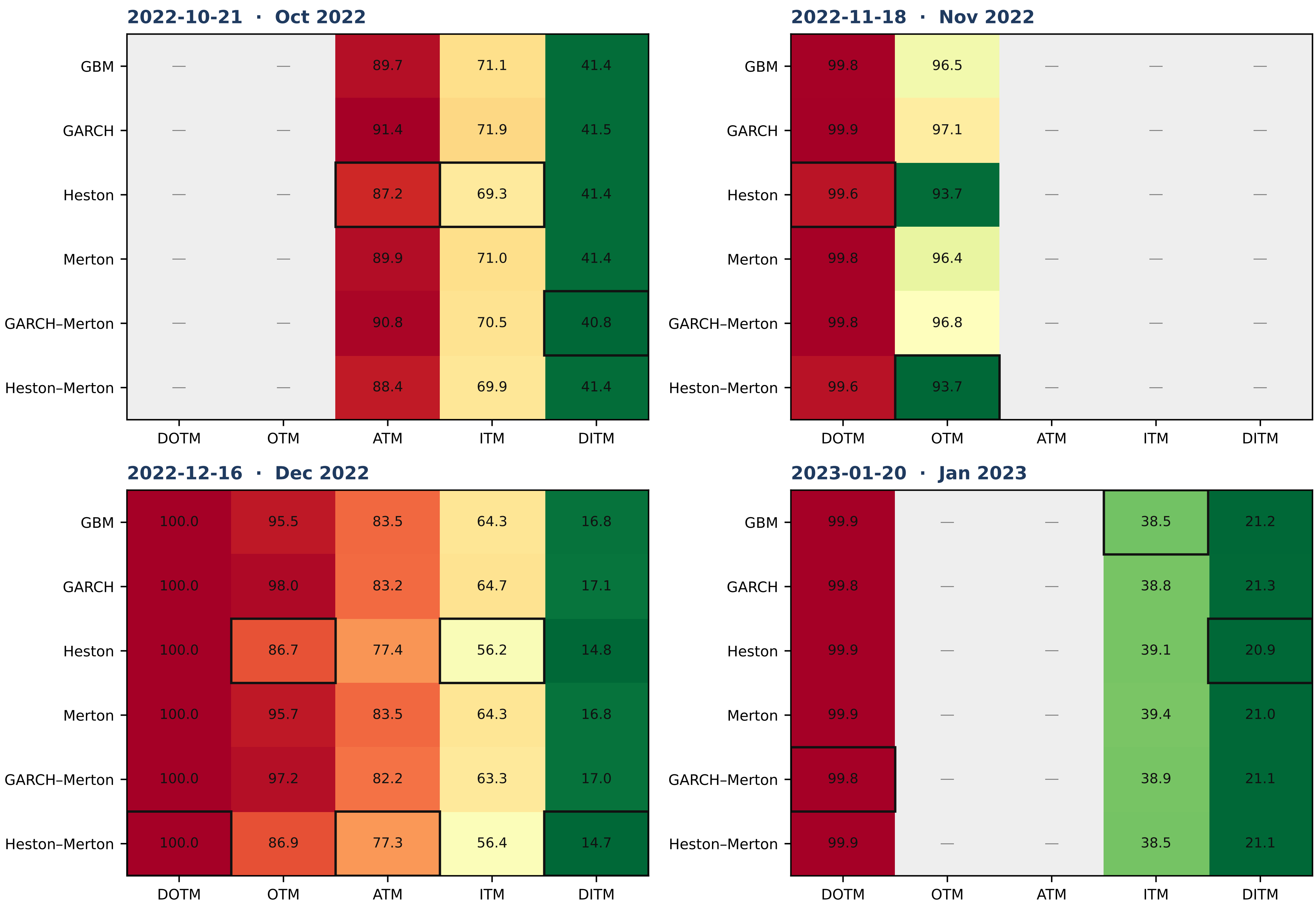
ITM/DITM: S > K. DOTM/OTM: S < K. ATM: |S/K − 1| ≤ 0.5%.

Page 5 · Figure M1 · Model × Moneyness RMSE% · overall (pooled)



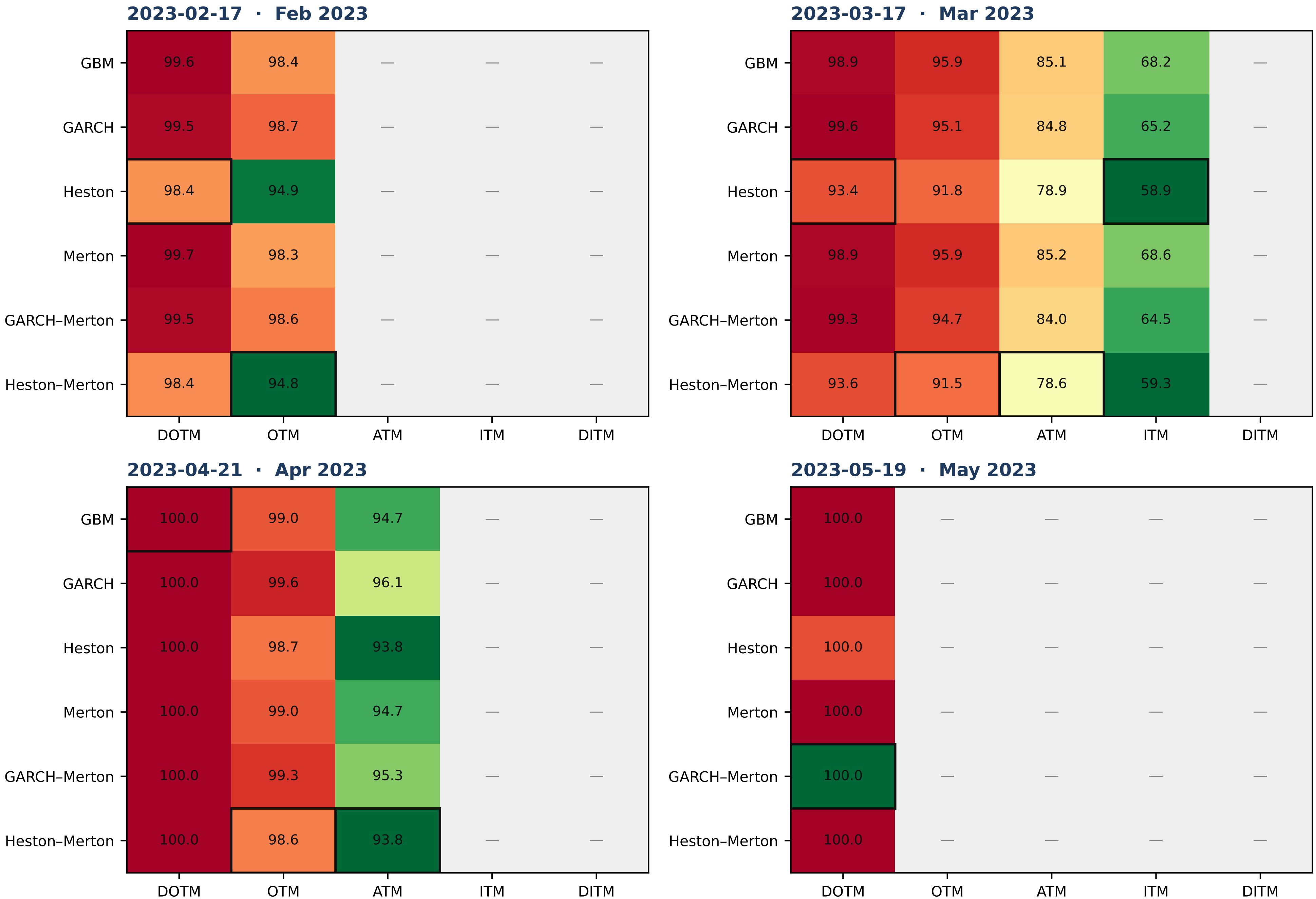
Black outline = lowest RMSE% in that moneyness column. Grey = n = 0. Pooled across companies and regimes.

Figure M2 · RMSE% by model × moneyness · each window (companies pooled)



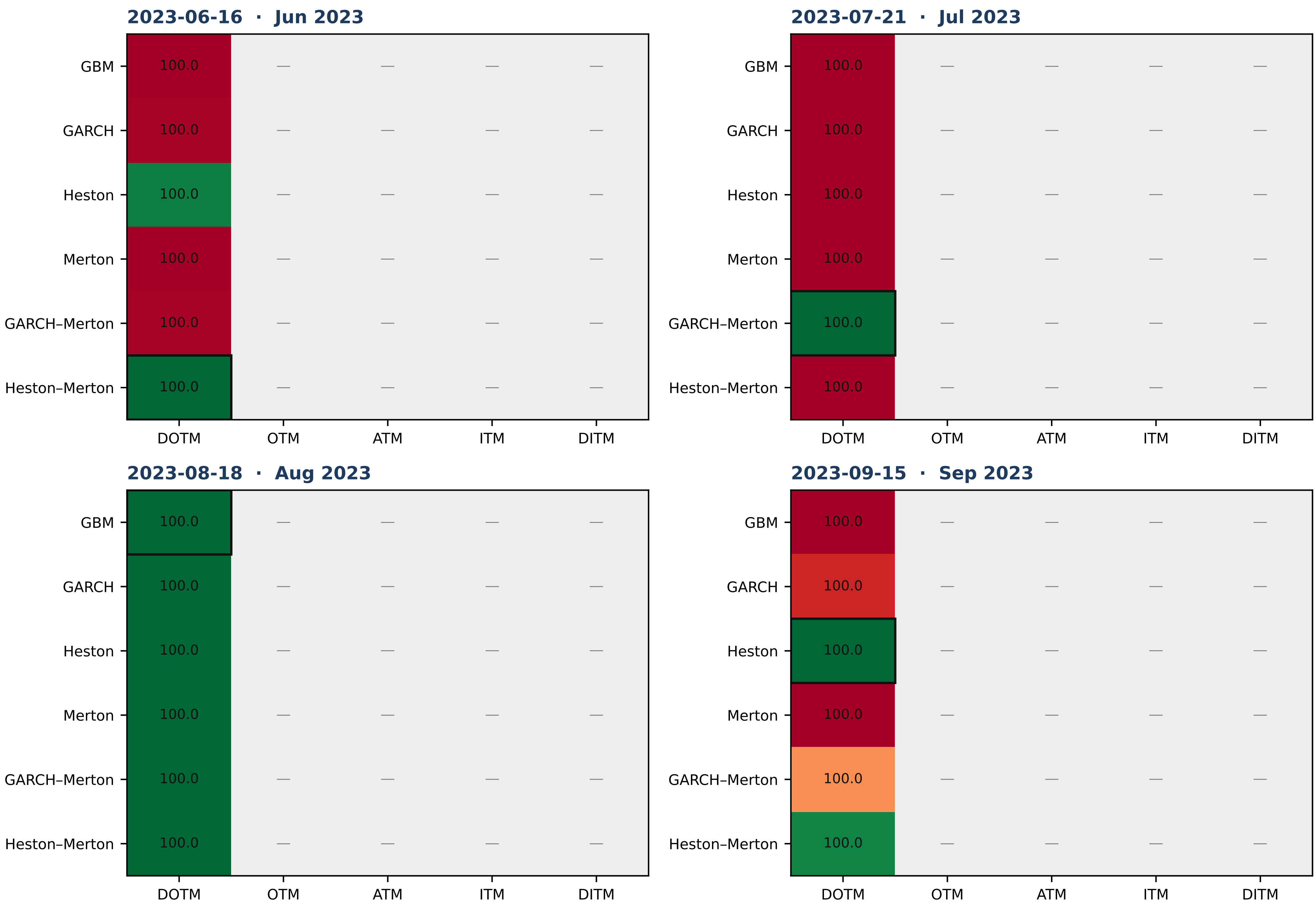
Outlined cell = lowest RMSE% in that window × moneyness column. Grey = n = 0.

Figure M2 · RMSE% by model × moneyness · each window (companies pooled)



Outlined cell = lowest RMSE% in that window × moneyness column. Grey = n = 0.

Figure M2 · RMSE% by model × moneyness · each window (companies pooled)



Outlined cell = lowest RMSE% in that window × moneyness column. Grey = n = 0.

Analysis 2 · Does the ranking change with moneyness inside each window?

Table M6 · Best model (lowest RMSE%) by window × moneyness

Window	DOTM	OTM	ATM	ITM	DITM
2022-10-21 Oct 2022	—	—	Heston	Heston	GARCH–Merton
2022-11-18 Nov 2022	Heston	Heston–Merton	—	—	—
2022-12-16 Dec 2022	Heston–Merton	Heston	Heston–Merton	Heston	Heston–Merton
2023-01-20 Jan 2023	GARCH–Merton	—	—	GBM	Heston
2023-02-17 Feb 2023	Heston	Heston–Merton	—	—	—
2023-03-17 Mar 2023	Heston	Heston–Merton	Heston–Merton	Heston	—
2023-04-21 Apr 2023	GBM	Heston–Merton	Heston–Merton	—	—
2023-05-19 May 2023	GARCH–Merton	—	—	—	—
2023-06-16 Jun 2023	Heston–Merton	—	—	—	—
2023-07-21 Jul 2023	GARCH–Merton	—	—	—	—
2023-08-18 Aug 2023	GBM	—	—	—	—
2023-09-15 Sep 2023	Heston	—	—	—	—

Table M7 · n contracts by window × moneyness (shared across models)

Window	DOTM	OTM	ATM	ITM	DITM
2022-10-21 Oct 2022	0	0	1	5	12
2022-11-18 Nov 2022	13	5	0	0	0
2022-12-16 Dec 2022	6	1	3	3	5
2023-01-20 Jan 2023	12	0	0	1	5
2023-02-17 Feb 2023	12	6	0	0	0
2023-03-17 Mar 2023	2	8	7	1	0
2023-04-21 Apr 2023	5	9	4	0	0
2023-05-19 May 2023	18	0	0	0	0
2023-06-16 Jun 2023	18	0	0	0	0
2023-07-21 Jul 2023	18	0	0	0	0
2023-08-18 Aug 2023	18	0	0	0	0
2023-09-15 Sep 2023	18	0	0	0	0

A change in the Table M6 entry across a row means the ranking depends on moneyness inside that window. Columns with n = 0 cannot change the ranking.

Page 8 · Findings

Does performance change with moneyness?

- DOTM (n = 140): BEST = Heston with RMSE% = 99.72, Bias% = -99.71.
- OTM (n = 29): BEST = Heston–Merton with RMSE% = 94.63, Bias% = -94.41.
- ATM (n = 15): BEST = Heston–Merton with RMSE% = 83.35, Bias% = -82.71.
- ITM (n = 10): BEST = Heston with RMSE% = 62.06, Bias% = -60.05.
- DITM (n = 22): BEST = GARCH–Merton with RMSE% = 32.77, Bias% = -28.54.
- Overall, the ranking does change with moneyness: DOTM → Heston, OTM → Heston–Merton, ATM → Heston–Merton, ITM → Heston, DITM → GARCH–Merton.
- Inside regimes, the BEST model also changes with moneyness in: 2022-10-21, 2022-11-18, 2022-12-16, 2023-01-20, 2023-02-17, 2023-03-17, 2023-04-21.
- These comparisons use the hourly nearest-ATM sample (09:59–15:59; LSM skipped at Friday 15:59). ATM still holds most of the mass; sparse DOTM/DITM buckets should be read as descriptive.